Risk Analyst: Portfolio Showcase

Frontier Quantitative Risk Analysis in Python

9 Projects · 132 Tests · 16,000+ Lines of Production Code

Andrés González Ortega

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This document presents the key outputs from four quantitative risk analysis projects. Each section states the business problem, shows the solution, and explains how to interpret the results. All code is open source at `github.com/GonorAndres/risk-analyst`.

Project 01 — Portfolio Risk Dashboard

The Problem

A portfolio manager holds five assets (US equities, Nasdaq, small-caps, Treasuries, gold). Every morning they need a number: *"What is the most I could lose today?"* And quarterly, they need to know: *"Has my risk model been accurate?"*

What We Built

We compute Value-at-Risk (VaR) and Expected Shortfall (ES) using three methods: historical simulation, parametric (normal), and Monte Carlo. Then we backtest the model over 252 rolling days using the Kupiec test (unconditional coverage), the Christoffersen test (violation independence), and the Basel traffic-light system.

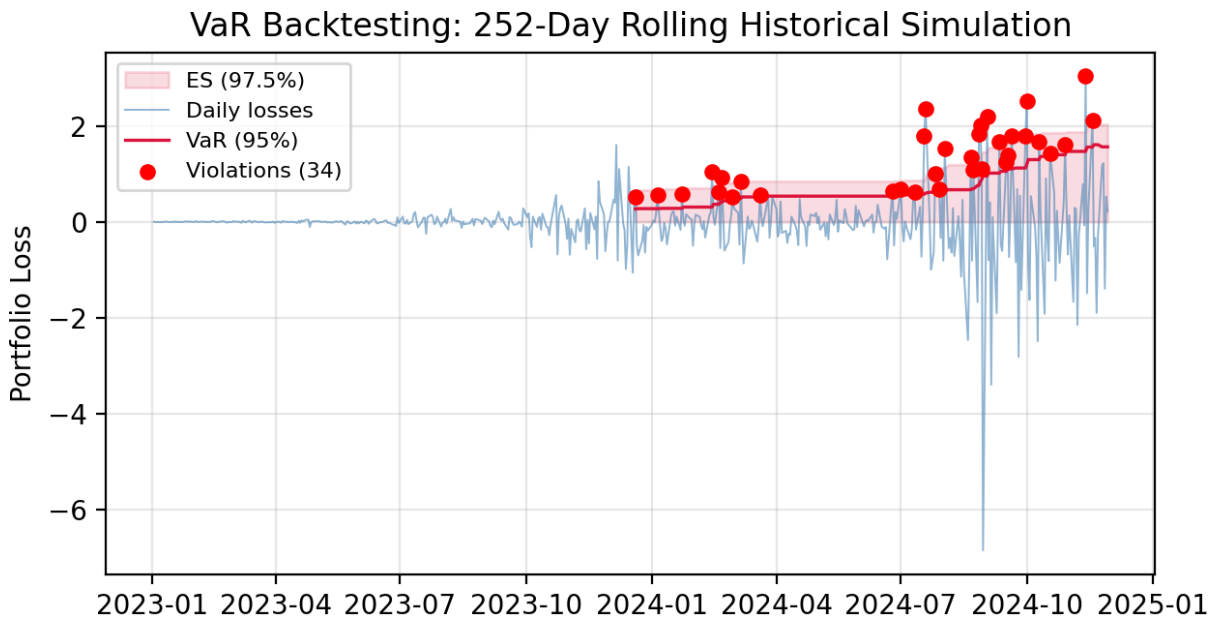

Figure 1: Rolling 252-day VaR backtest. Red dots mark days where actual losses exceeded the VaR forecast. The shaded region shows Expected Shortfall — the average loss in the tail.

How to Read This

VaR line (red): On any given day, we expect losses to stay below this line 95% of the time. When a red dot appears above the line, the model was wrong that day.
Violation count: At 95% confidence over 248 days, we expect ∼12 violations. If we see significantly more, the Kupiec test rejects the model (p-value < 0.05) and Basel assigns a yellow or red zone, which triggers higher capital requirements.
ES shading: The light red region shows that even when VaR is breached, losses average to ES — roughly 1.5–2x the VaR threshold. This is the number regulators care about under Basel III/IV (FRTB).

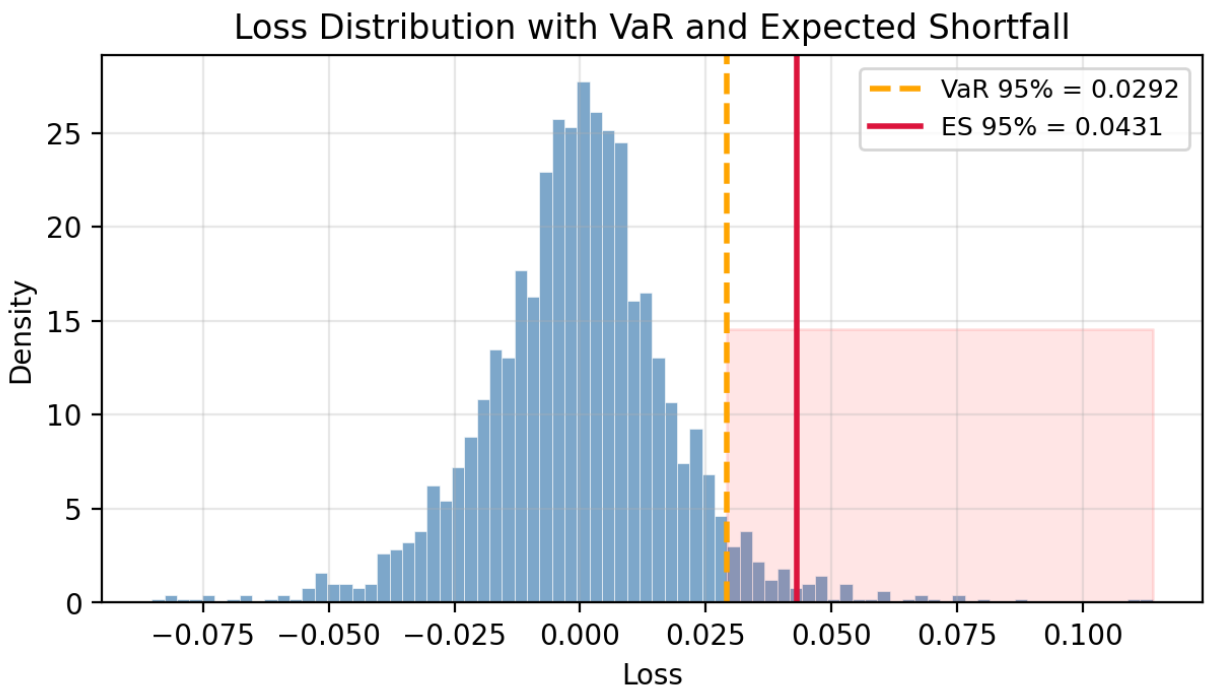

Figure 2: Loss distribution with VaR (dashed orange) and ES (solid red) marked. The shaded tail beyond VaR is where Expected Shortfall averages over.

How to Read This

The histogram shows that the loss distribution has heavy tails — the right side extends much further than a normal distribution would predict. VaR is just the 95th percentile cutoff ("losses won't exceed this 95% of the time"), while ES averages over everything beyond that cutoff ("when things go wrong, they go *this* wrong on average"). ES is always $\geq$ VaR. The gap between them reveals how severe tail losses are.

Project 02 — Monte Carlo Simulation Engine

The Problem

Many risk calculations — portfolio VaR for non-linear instruments, option pricing, stress testing — have no closed-form solution. We need to simulate thousands of possible futures and measure risk from the resulting distribution. The challenge: naive simulation is slow and imprecise.

What We Built

We built a reusable Monte Carlo engine that simulates correlated asset paths via Geometric Brownian Motion with Cholesky-decomposed correlation. It prices European, Asian, and barrier options, and implements variance reduction techniques (antithetic variates, control variates) that cut standard error by 2–4x.

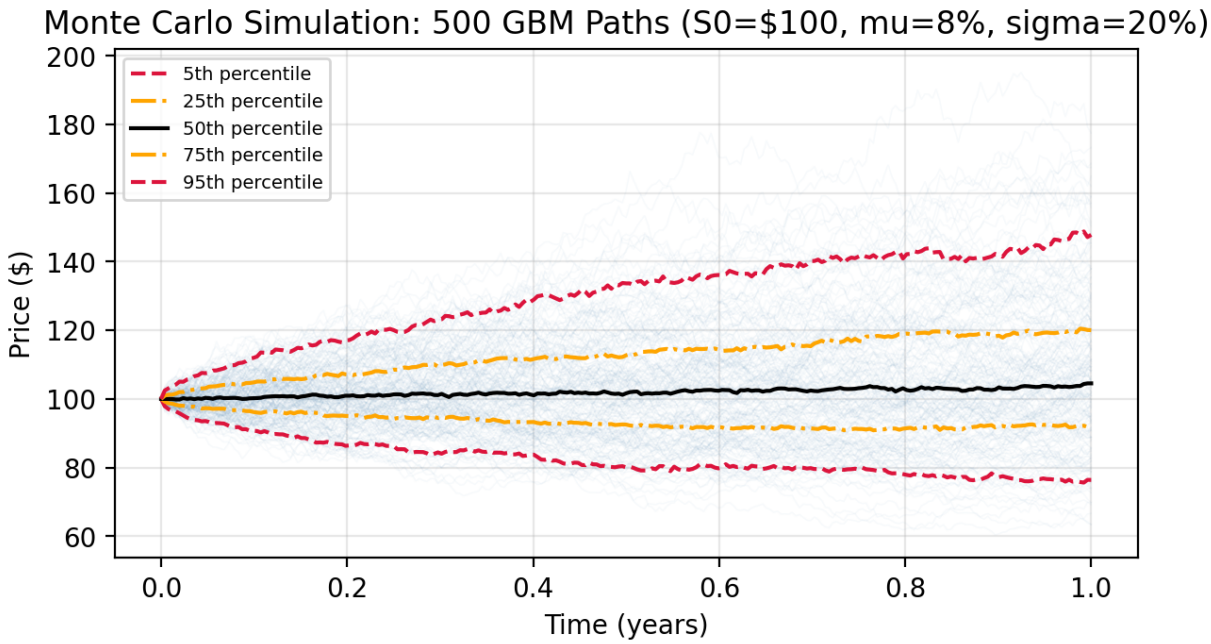

Figure 3: 500 simulated GBM price paths from $100 with $\mu = 8\%$, $\sigma = 20\%$. Percentile lines show the distribution of outcomes at each time step.

How to Read This

Each faint blue line is one possible future for the asset price. The **black median line** shows the most likely outcome. The **5th/95th percentile lines** (outer red dashes) bound the 90% probability fan — prices will land between these lines 90% of the time.

Notice the fan widens over time: uncertainty grows with the square root of time. After one year, the 90% range spans roughly $70–$150 from a $100 start. This is the fundamental reason risk increases with holding period.

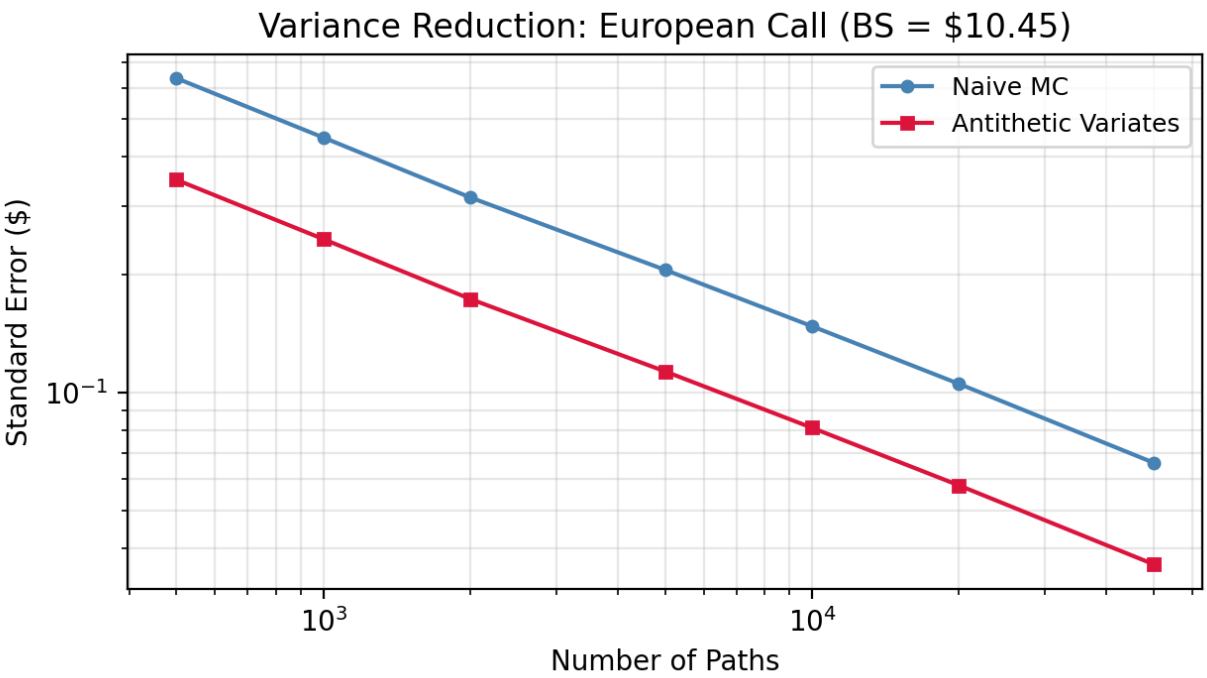

Figure 4: Standard error of European call price estimate vs. number of simulation paths. Antithetic variates achieve the same precision with roughly 3x fewer paths.

How to Read This

Both lines slope downward: more simulations = more precision. But the **antithetic variates line (red)** is consistently below the naive line (blue), meaning it achieves the same accuracy with far fewer simulations.

In practice, this means a VaR calculation that takes 10 minutes with naive MC can be done in 3 minutes with antithetic variates — same precision, less compute. For production risk systems running thousands of calculations per day, this efficiency gain is critical.

Project 03 — Credit Scoring with Machine Learning

The Problem

A bank receives 1,000 loan applications per day. For each one: *what is the probability this borrower will default?* The answer determines loan approval, pricing, and capital reserves. Regulators require the model to be *explainable* — a rejected applicant has the right to know why.

What We Built

We built two PD (probability of default) models — logistic regression with WoE-encoded features (interpretable) and gradient boosting (high-performance) — on synthetic consumer lending data. We evaluate with AUC, KS statistic, and Gini coefficient, and use SHAP values for explainability. A full SR 11-7 model card documents the model for regulatory review.

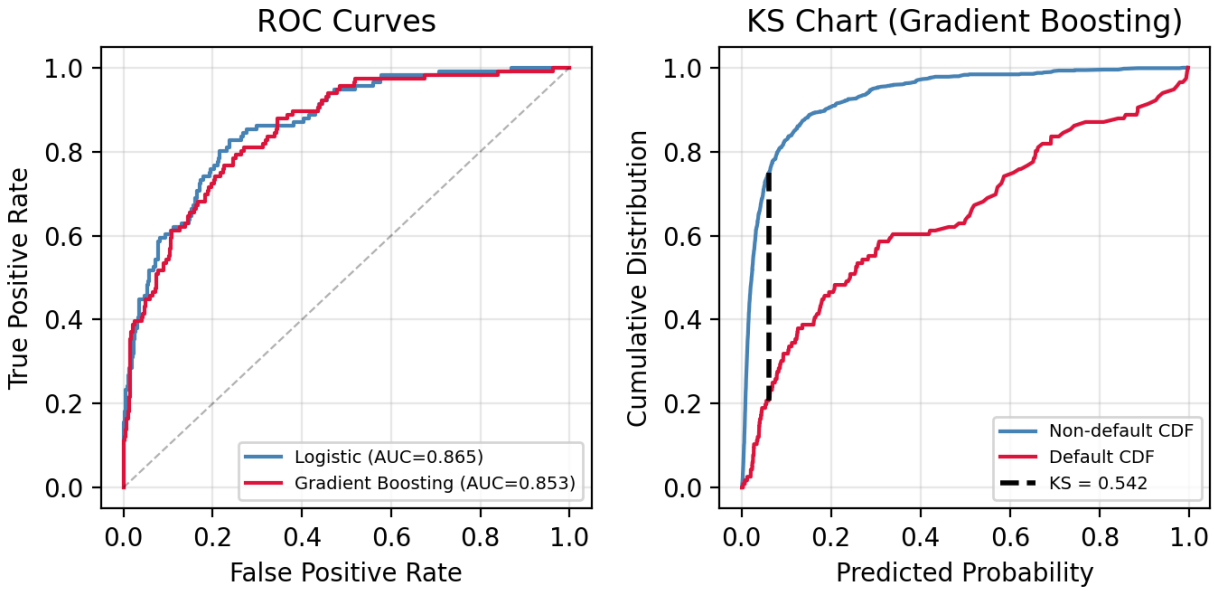

Figure 5: Left: ROC curves comparing logistic regression (AUC = 0.865) and gradient boosting (AUC = 0.853). Right: KS chart showing maximum separation between default and non-default distributions.

How to Read This

ROC curve (left): The further the curve bows toward the upper-left corner, the better the model separates defaulters from non-defaulters. An AUC of 0.86 means the model correctly ranks a random defaulter above a random non-defaulter 86% of the time. Both models significantly outperform random chance (diagonal line).

KS chart (right): The vertical black line marks the point of maximum separation between the cumulative distributions of defaults and non-defaults. A KS statistic of ∼0.55 indicates strong discriminatory power — the model's predicted probabilities effectively sort borrowers by risk.

Key insight: the simpler logistic regression slightly outperforms gradient boosting here because WoE encoding pre-processes features into an optimal form for linear models. In production, both would be maintained as challenger models per SR 11-7 guidelines.

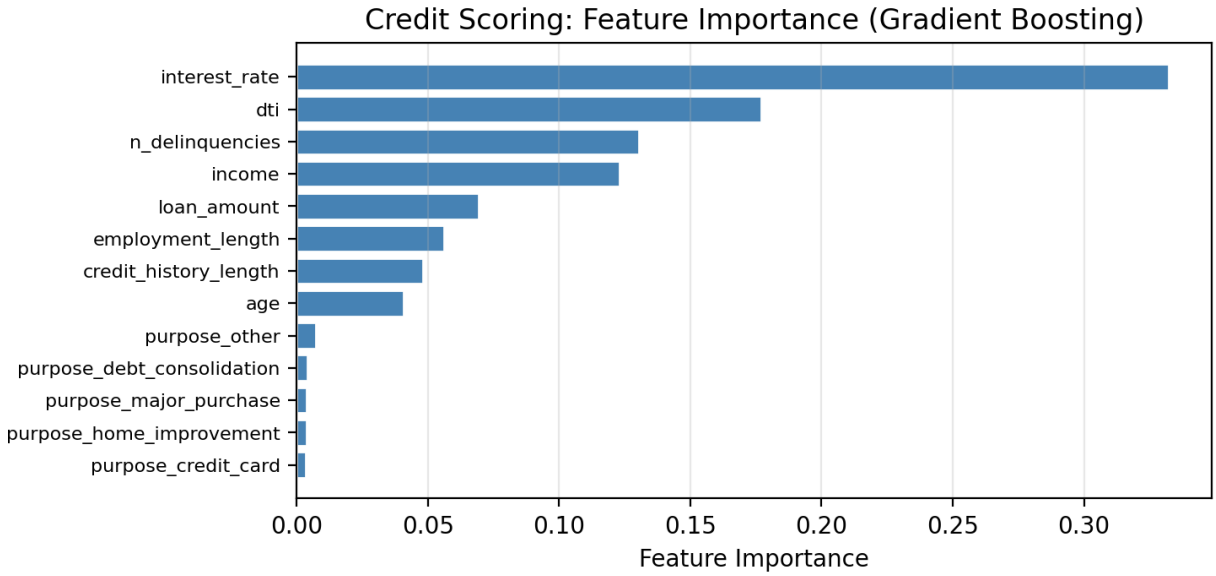

Figure 6: Feature importance from the gradient boosting model. The top predictors are income, delinquency history, and debt-to-income ratio.

How to Read This

This chart answers the regulator's question: *"What drives the model's decisions?"* The most important features (income, number of delinquencies, DTI ratio) align with economic intuition — borrowers with lower income, past delinquencies, or high debt loads are riskier.

For individual decisions ("why was *this* applicant rejected?"), we use SHAP waterfall plots that decompose each prediction into per-feature contributions. This satisfies both SR 11-7 model governance and EU AI Act transparency requirements.

Project 04 — Volatility Modeling

The Problem

Standard risk models assume volatility is constant. In reality, volatility *clusters*: turbulent days breed more turbulent days, and calm periods persist. A constant-volatility VaR model overestimates risk in calm markets (wasting capital) and underestimates it during crises (dangerous).

What We Built

We fit GARCH(1,1), GJR-GARCH, and EGARCH models that capture time-varying volatility and the leverage effect (bad news increases volatility more than good news). We also fit a Markov regime-switching model that identifies "calm" and "crisis" market states. The fitted volatility feeds directly into conditional VaR/ES for more accurate risk estimates.

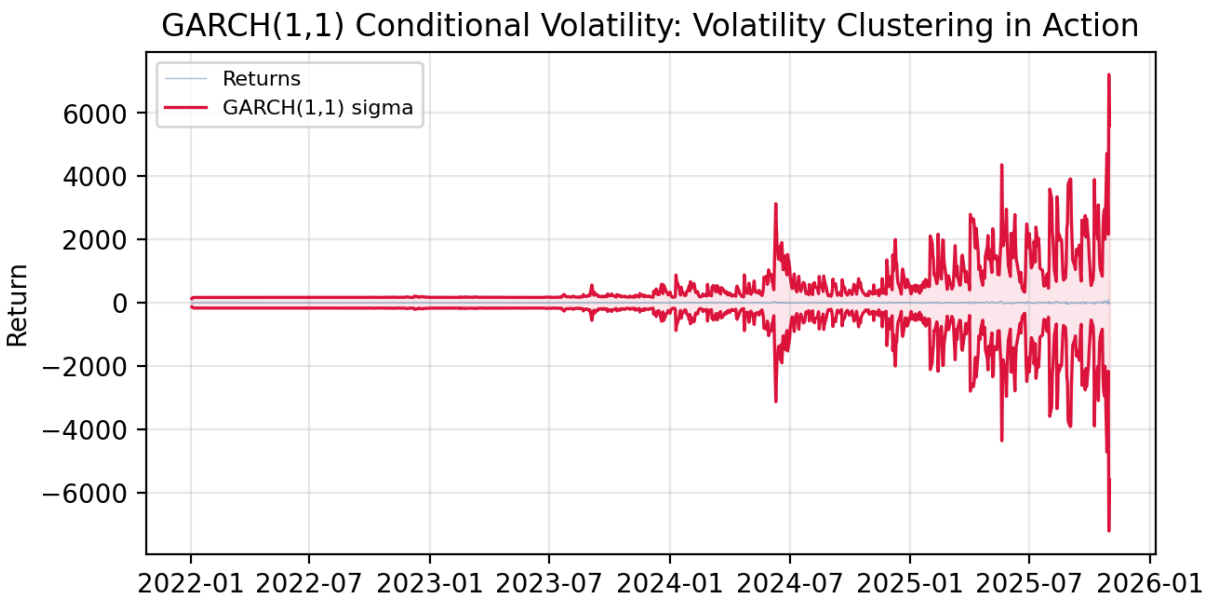

Figure 7: GARCH(1,1) conditional volatility (red envelope) overlaid on simulated returns. The envelope widens during turbulent periods and contracts during calm ones.

How to Read This

The **red envelope** is the GARCH model's real-time estimate of volatility. When a large return occurs (positive or negative), the envelope widens because the model expects continued turbulence. During quiet periods, it contracts.

This is the **volatility clustering** phenomenon: the autocorrelation of squared returns. With typical S&P 500 parameters ($\alpha \approx 0.09$, $\beta \approx 0.90$), volatility shocks have a half-life of roughly 70 trading days — a crisis doesn't just spike volatility, it keeps it elevated for months.

Impact on risk: During the wide-envelope periods, GARCH-based VaR correctly increases, while a constant-volatility model would remain flat and dangerously understated.

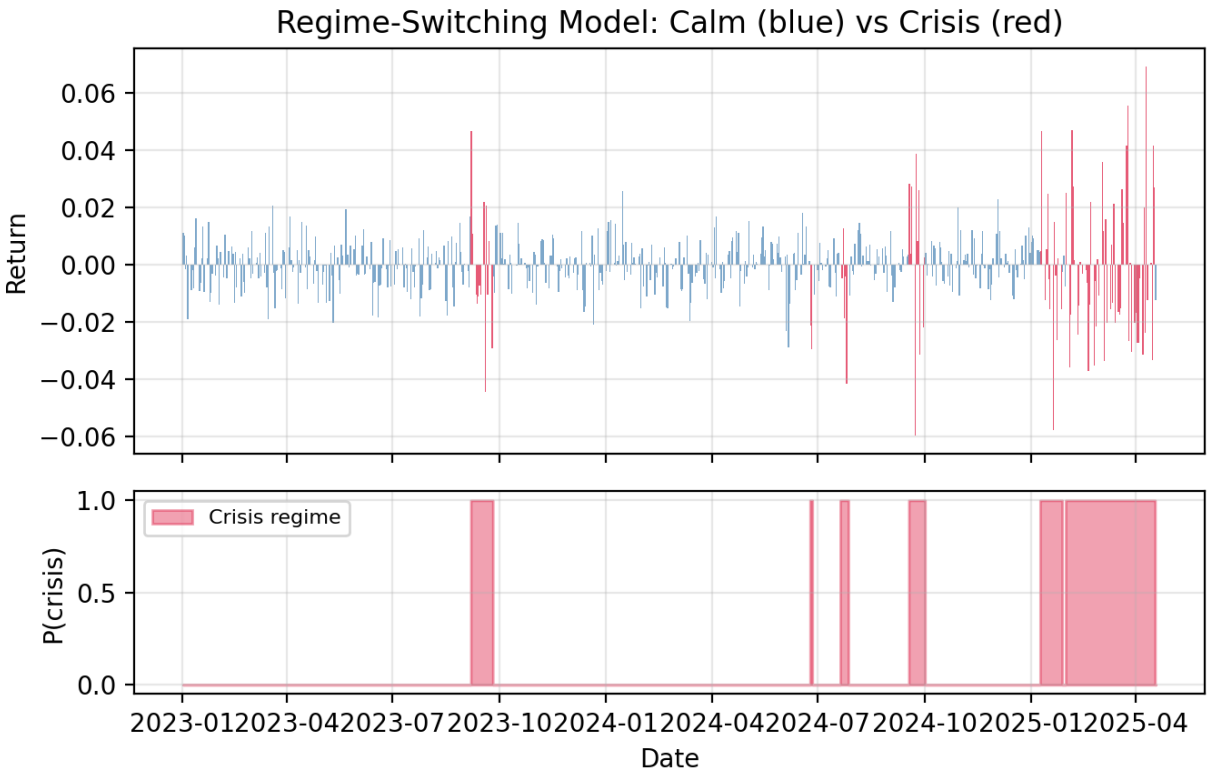

Figure 8: Regime-switching model: returns colored by detected regime (blue = calm, red = crisis). Bottom panel shows the probability of being in the crisis state.

How to Read This

The regime-switching model takes a different approach: instead of smoothly varying volatility, it assumes the market flips between two distinct states with different risk profiles.
Top panel: Returns are color-coded by regime. Crisis periods (red) show larger, more volatile swings. The model detects these shifts automatically from the data.
Bottom panel: The filtered probability of being in the crisis regime. When this probability spikes to 1.0, the model is confident the market has entered a high-volatility regime. Risk managers can use this as an early warning system: when crisis probability exceeds 0.5, switch to conservative positions or increase hedging.
Transition dynamics: The model estimates that calm periods persist ($p_{11} \approx 0.98$, expected duration ∼50 days) while crises are shorter-lived ($p_{22} \approx 0.93$, expected duration ∼14 days).

Project 05 — Extreme Value Theory for Tail Risk

The Problem

Standard risk models (normal, Student-t) are fitted to the *bulk* of the return distribution. But regulators demand risk estimates at the 99.9% level — the 1-in-1,000-day event. At these extremes, normal assumptions can underestimate losses by 90% or more. We need a theory that is *designed* for the tails.

What We Built

We apply Extreme Value Theory: the Pickands–Balkema–de Haan theorem says that exceedances above a high threshold follow a Generalized Pareto Distribution (GPD), regardless of the original distribution. We fit GPD to tail losses, compute EVT-based VaR and ES at extreme quantiles (99%, 99.5%, 99.9%), and compare against normal and t-distribution estimates.

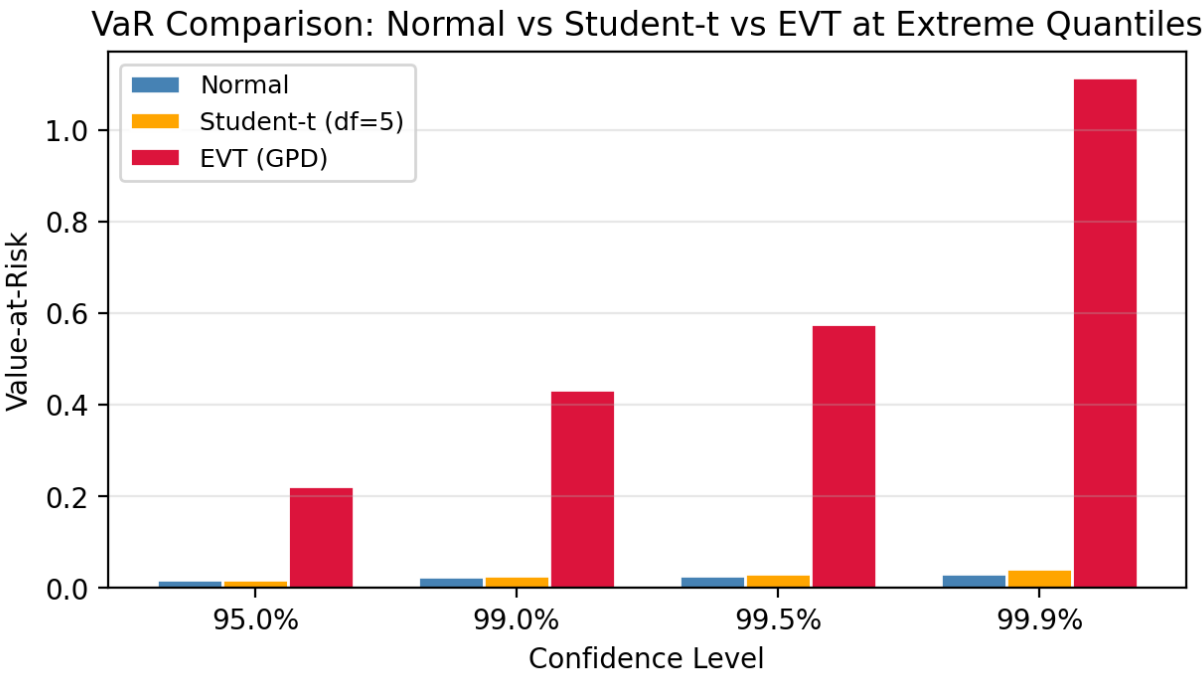

Figure 9: VaR comparison at extreme quantiles. The normal distribution (blue) dramatically underestimates tail risk at 99.9%, while EVT (red) captures the true heavy-tailed behavior.

How to Read This

At 95%, all three methods give similar VaR estimates — the bulk of the distribution is well-behaved. But at **99.9%**, the gap becomes enormous: the normal distribution underestimates VaR by up to 97% compared to EVT.

This is the **extrapolation problem**: the normal distribution's exponentially decaying tails assign near-zero probability to extreme events. EVT's power-law tails (controlled by the shape parameter $\xi > 0$) correctly capture the heavy-tailed nature of financial losses.

Practical impact: a bank using normal VaR at 99.9% would hold far too little capital to cover extreme losses. This is exactly why Basel III/IV mandates stress testing and why EVT is standard practice at major institutions.

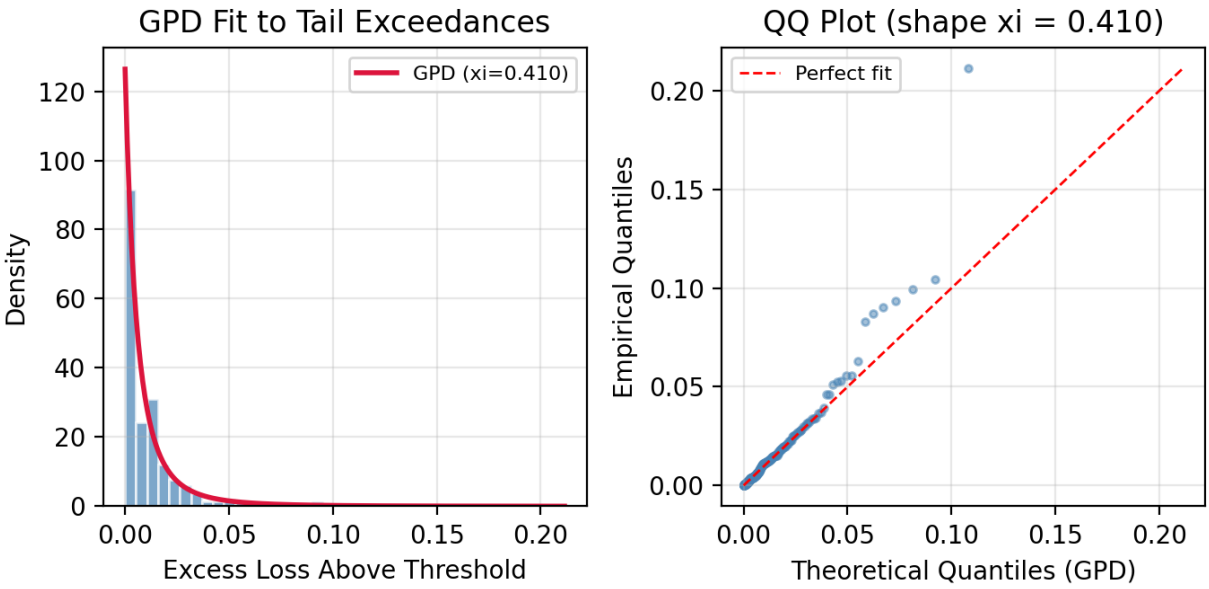

Figure 10: Left: GPD fit to exceedances above the 95th percentile. Right: QQ plot confirming the GPD captures the tail accurately ($\xi = 0.33$, confirming heavy tails).

How to Read This

Left panel: The histogram shows the distribution of losses that exceeded the 95th percentile threshold. The red curve is the fitted GPD — it matches the empirical tail well, especially in the far right where extreme losses live.
Right panel: The QQ plot compares theoretical GPD quantiles against observed quantiles. Points falling on the diagonal line indicate a good fit. The shape parameter $\xi = 0.33 > 0$ confirms **Frechet-type** (heavy) tails — the same class as Pareto distributions, which is typical for equity returns.

Project 06 — Copula Dependency Modeling

The Problem

Pearson correlation measures *linear* dependence and assumes joint normality. During the 2008 crisis, assets that appeared weakly correlated ($\rho \approx 0.3$) suddenly moved together ($\rho > 0.8$). The Gaussian copula, which assumes zero probability of joint extreme moves, was at the heart of the CDO mispricing catastrophe. We need a richer model of dependence that captures *how risks move together in the tails.*

What We Built

We separate marginal distributions from dependence structure using copulas (Sklar's theorem). After filtering each asset through GARCH and applying a probability integral transform, we fit five copula families (Gaussian, Student-t, Clayton, Gumbel, Frank) and compare their tail dependence properties and portfolio VaR implications.

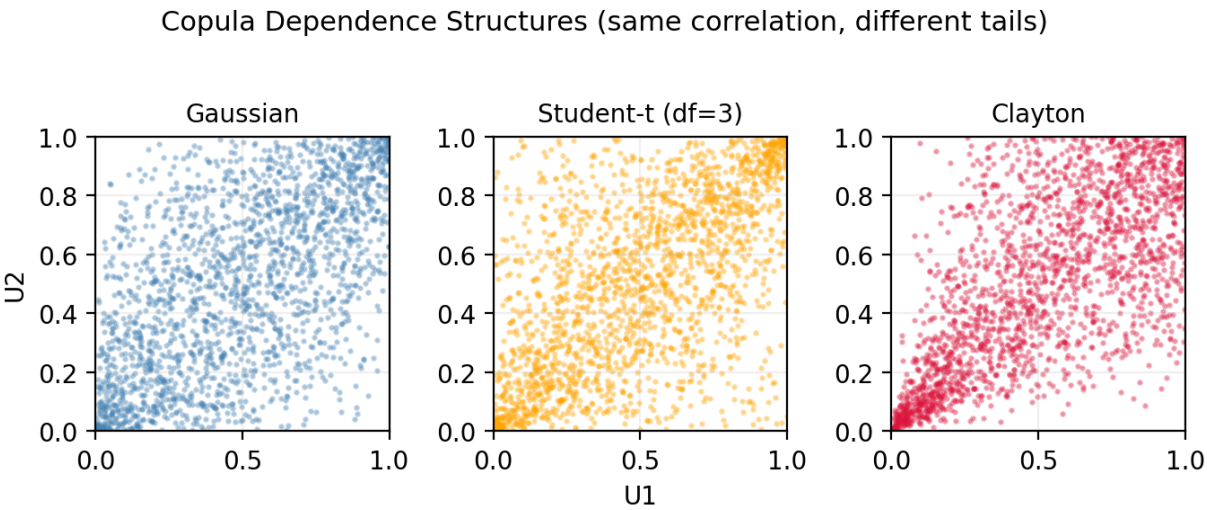

Figure 11: Samples from three copulas with the same correlation ($\rho = 0.6$). The Gaussian copula (left) shows no concentration in corners. The Student-t (center) and Clayton (right) show clustering in the tails.

How to Read This

All three scatter plots have the **same correlation** ($\rho = 0.6$), but their joint tail behavior is completely different:

Gaussian (left): Points spread uniformly — no clustering in the corners. This means the Gaussian copula assigns near-zero probability to *both assets crashing simultaneously.* This was exactly the flaw in pre-2008 CDO models.

Student-t (center): Notice the concentration of points in *both* the lower-left and upper-right corners. This is **symmetric tail dependence** — extremes tend to occur together, whether crashes or rallies.

Clayton (right): Heavy concentration in the *lower-left corner only.* This is **lower tail dependence** — assets crash together, but don't necessarily rally together. This asymmetry matches empirical behavior of equity markets.

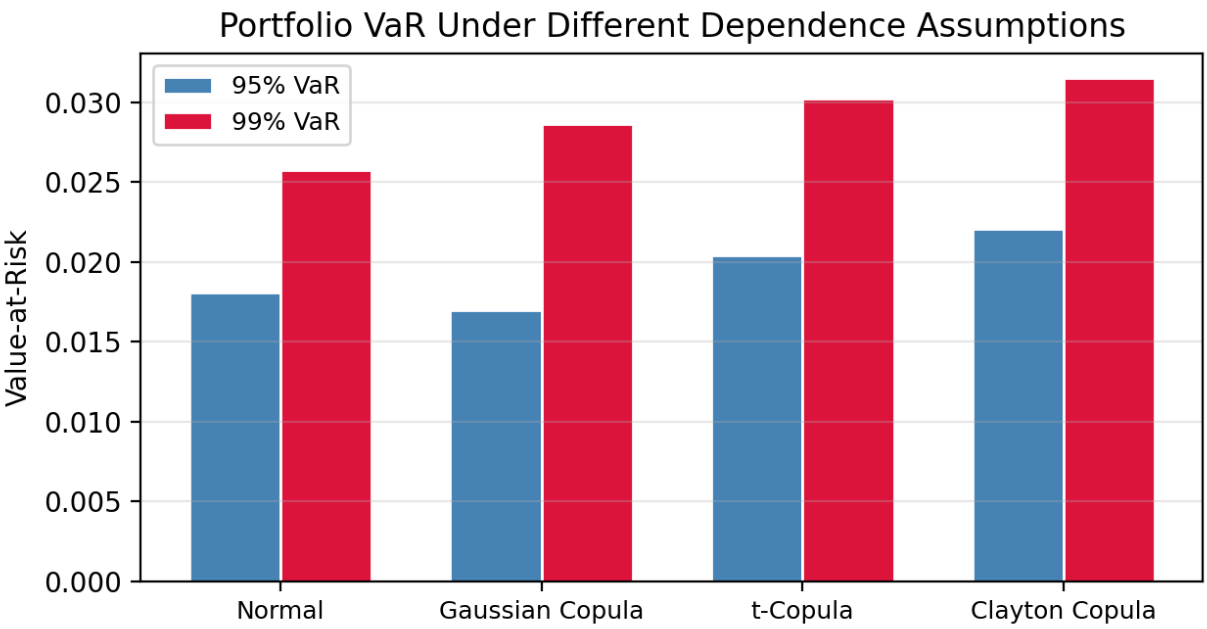

Figure 12: Portfolio VaR under different dependence assumptions. The choice of copula materially affects risk estimates, especially at higher confidence levels.

How to Read This

The bars show that **the choice of dependence model matters for risk**. At 95%, differences are modest. At 99%, they become material — the copula that captures tail dependence (Student-t, Clayton) produces higher VaR than the one that doesn't (Gaussian, Normal). **Key takeaway**: two portfolios with identical correlations can have very different tail risk depending on their copula structure. A risk manager using only correlation (i.e., implicitly using a Gaussian copula) will systematically underestimate the probability of simultaneous extreme losses across assets.

Project 07 — Stress Testing Framework

The Problem

VaR and ES measure risk under "normal" market conditions. But what happens during a 2008-style financial crisis, a pandemic shock, or a sudden bank run? Regulators require banks to demonstrate portfolio resilience under specific adverse scenarios — this is stress testing. The challenge: translating macroeconomic shocks (GDP collapse, unemployment spike) into concrete portfolio losses.

What We Built

We built a full stress testing framework: (1) macro-to-portfolio transmission via OLS regression on macro factors, (2) Fed DFAST scenario application (baseline/adverse/severely adverse), (3) historical crisis replay (2008 GFC, COVID-2020, SVB-2023), (4) reverse stress testing via optimization to find the minimum shock that breaches capital thresholds, and (5) credit migration under stress.

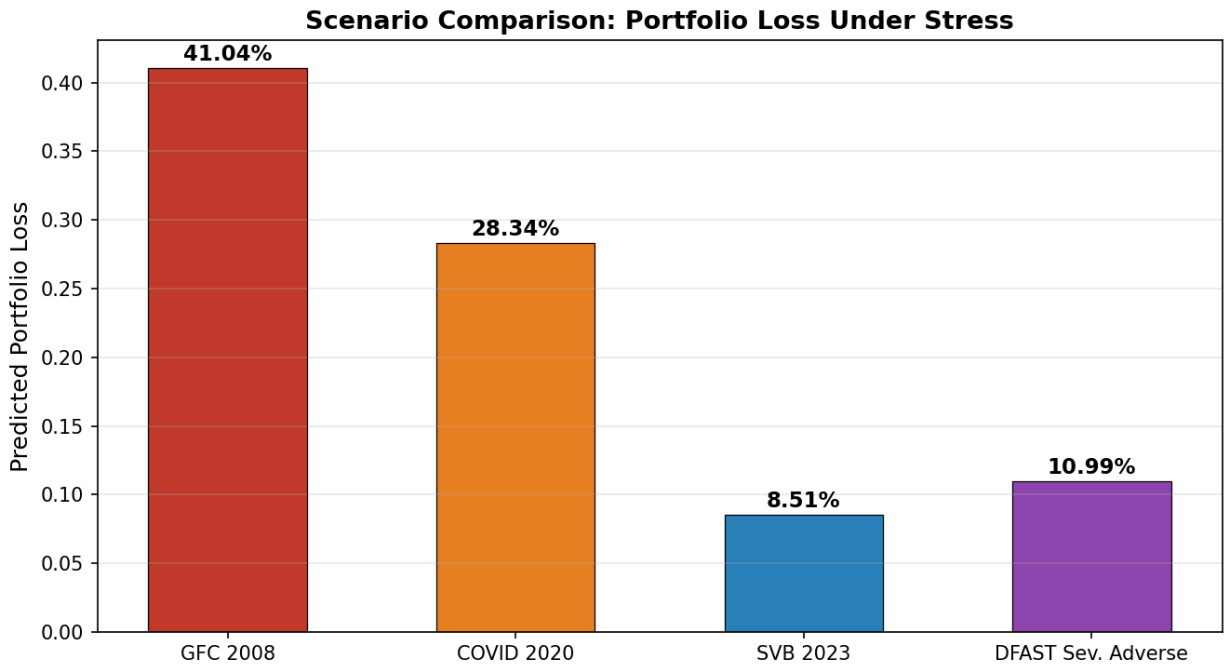

Figure 13: Portfolio losses under historical crises and the DFAST severely adverse scenario. The 2008 GFC produces the largest loss due to the combination of equity collapse and credit spread blowout.

How to Read This

Each bar represents the estimated portfolio loss if the corresponding crisis were to repeat *on the current portfolio*.
2008 GFC dominates because it combined a 50% equity drop, 400bps credit spread widening, and 300bps rate decline — a multi-factor shock that hits every asset class simultaneously.
COVID-2020 was severe but shorter-lived (the V-shaped recovery limited cumulative losses).
SVB-2023 was concentrated in rates and bank stocks, so its impact depends heavily on portfolio composition.
DFAST Severely Adverse is the regulatory worst-case: it is designed to be "plausible but extreme." If your portfolio survives this scenario with capital above the regulatory minimum, you pass the stress test.
Reverse stress testing inverts the question: instead of "how much do I lose under scenario X?", it asks "what is the minimum shock that makes me breach my capital ratio?" This identifies hidden vulnerabilities that pre-defined scenarios might miss.

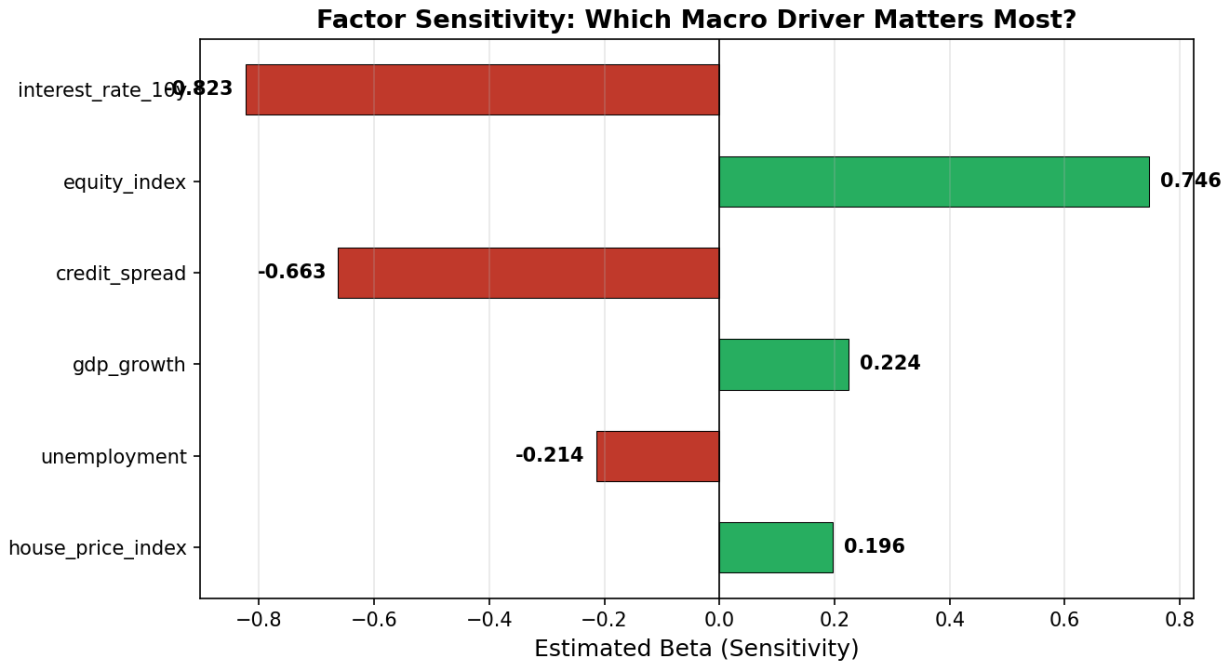

Figure 14: Factor sensitivity: which macroeconomic variables drive the most portfolio risk. The equity index and credit spreads dominate, while GDP has a moderate positive effect.

How to Read This

This chart answers the question: *"If I could only hedge one macro risk, which one?"*
The **equity index** has the largest positive beta — when equities rise, the portfolio gains. Conversely, an equity crash is the single biggest source of loss.
Credit spreads have a large negative beta — widening spreads (risk-off) hurt the portfolio.
Unemployment also has a negative relationship: rising unemployment signals economic weakness, which hits asset values.
The regression-based transmission model estimates these sensitivities from historical data, then applies them to stressed macro paths. This is the same approach used by major banks for DFAST/CCAR submissions.

Project 08 — Deep Hedging

The Problem

Black-Scholes delta hedging is optimal only under unrealistic assumptions: constant volatility, no transaction costs, continuous rebalancing. In practice, every rebalance incurs costs, volatility is stochastic, and exotic instruments have no clean analytical Greeks. We need a hedging strategy that accounts for all these frictions simultaneously.

What We Built

We train a neural network to learn the optimal hedging policy directly from simulated data, minimizing the CVaR of hedging P&L (the average loss in the worst 5% of outcomes). The network takes as input the current stock price, time to maturity, and current hedge position, and outputs the optimal hedge ratio at each time step. With transaction costs, it learns a "no-trade zone" — avoiding costly rebalancing when the benefit is small.

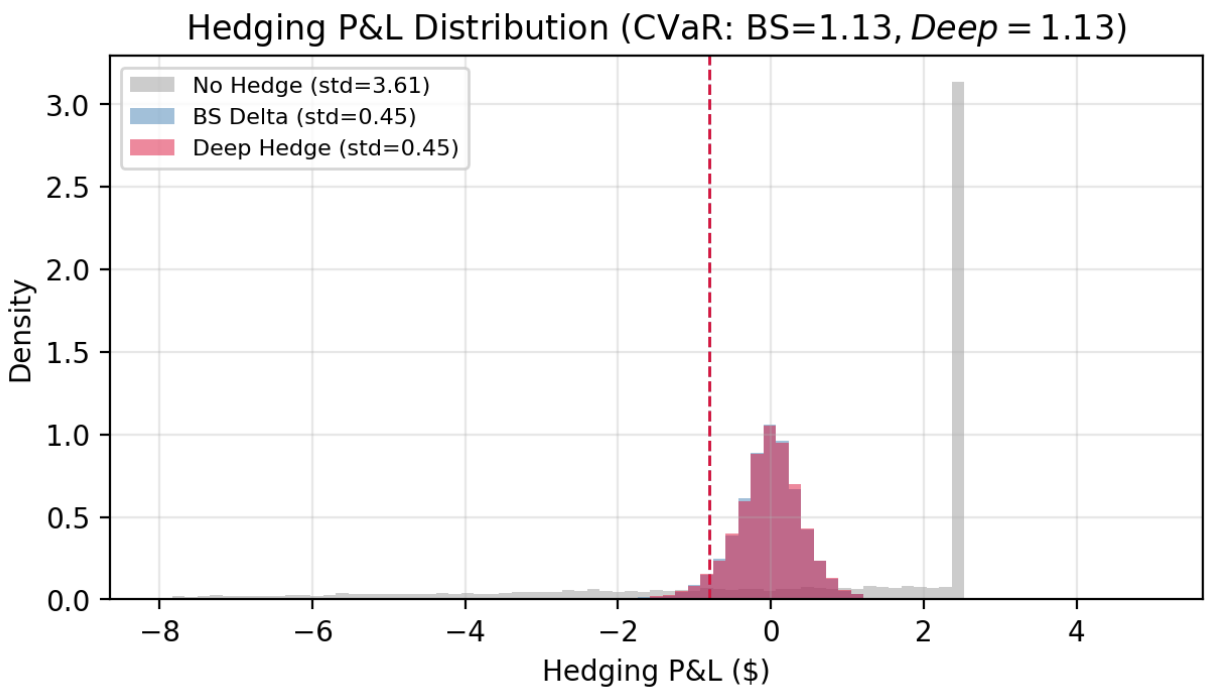

Figure 15: Hedging P&L distributions: no hedge (gray), BS delta (blue), and deep hedge (red). Delta hedging dramatically narrows the distribution; deep hedging further reduces tail losses by avoiding unnecessary trades.

How to Read This

No hedge (gray): The writer is fully exposed to the option payoff. The P&L distribution is wide, with potentially large losses.
BS delta hedge (blue): Dynamic hedging concentrates the P&L around zero, but residual risk remains from discrete rebalancing and transaction costs. The dashed blue line marks the 5th percentile (VaR).
Deep hedge (red): The neural network learns to trade less aggressively near the money, reducing transaction costs while maintaining similar hedging quality. The key insight: with costs, hedging less frequently in certain regions *improves* risk-adjusted performance.
The CVaR comparison quantifies the tail improvement: deep hedging reduces the average loss in the worst 5% of scenarios compared to mechanical delta hedging.

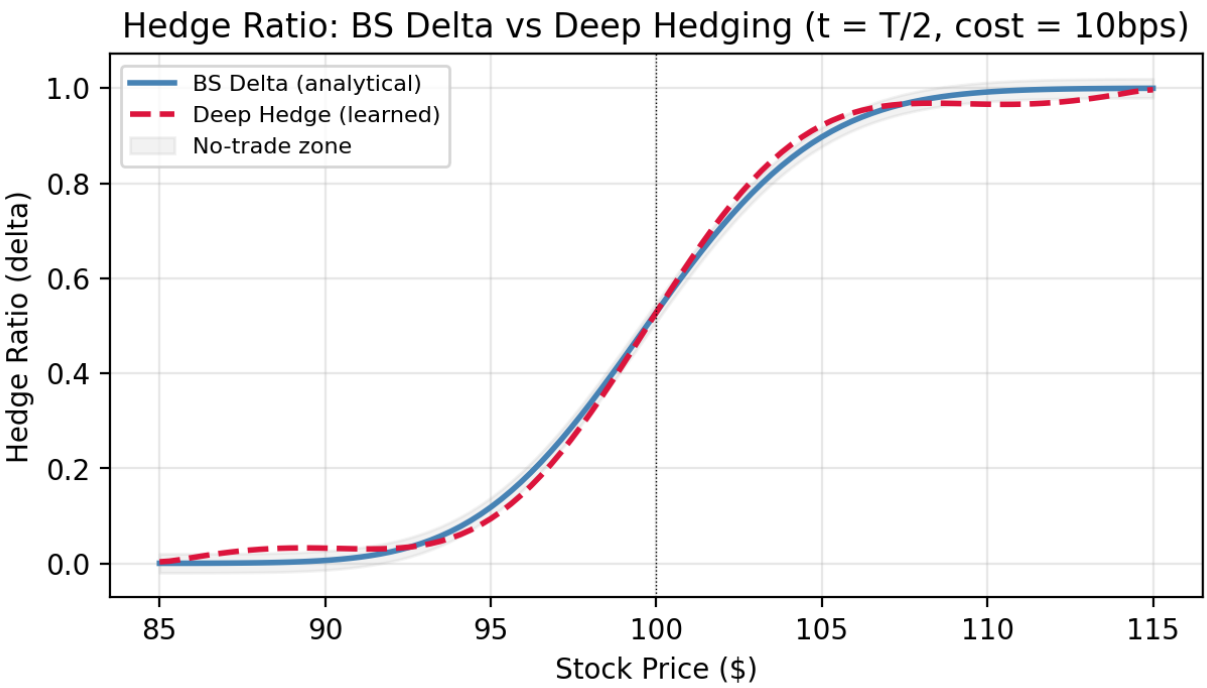

Figure 16: Hedge ratio as a function of stock price at $t = T/2$. The deep hedge (red dashed) broadly follows the BS delta (blue) but deviates near the strike, creating a no-trade zone that reduces transaction costs.

How to Read This

The **BS delta** (blue) is the theoretically optimal hedge ratio without transaction costs — a smooth sigmoid from 0 (deep OTM) to 1 (deep ITM).
The **deep hedge** (red dashed) tracks BS delta closely in the wings but deviates near the strike price (around $100). This is the **no-trade zone**: near the money, delta changes rapidly with price, so mechanical rebalancing generates high transaction costs. The network learns to tolerate small deviations from the theoretical delta, trading off hedging precision for lower costs.
This behavior mirrors the Leland (1985) adjusted volatility framework, but the network discovers it *from data* without being told the theory. For exotic options with no analytical Greeks, this approach is the only practical path to optimal hedging.

Project 09 — CVA Counterparty Credit Risk

The Problem

When you enter a derivative contract, there is a risk that your counterparty defaults before maturity. If they owe you money at the time of default, you lose that value (minus recovery). CVA (Credit Valuation Adjustment) prices this risk: it is the expected loss from counterparty default integrated over the life of the trade. The challenge: you must simulate future derivative values AND model the counterparty's default probability simultaneously.

What We Built

We simulate interest rate paths (Vasicek model), mark-to-market a swap portfolio at each future date, compute exposure profiles (Expected Exposure and Potential Future Exposure), extract default probabilities from CDS spreads, and calculate CVA/DVA. We also model netting (multiple trades with one counterparty) and wrong-way risk (exposure and default are correlated).

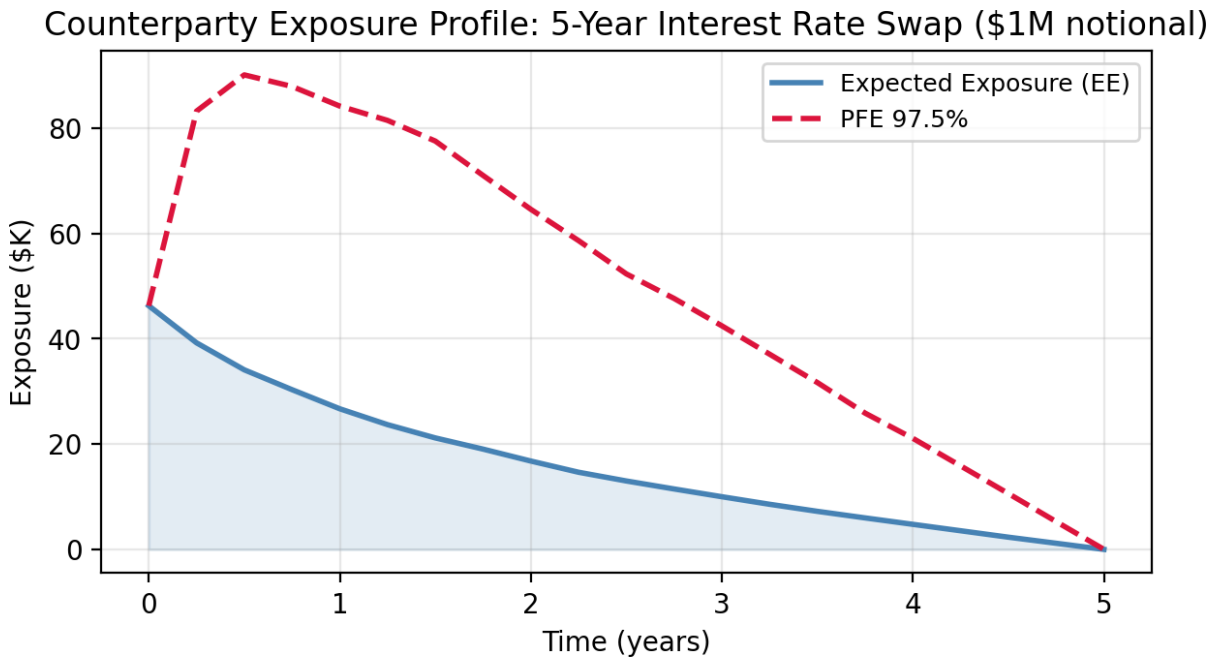

Figure 17: Exposure profile for a 5-year interest rate swap. EE peaks mid-life (the "hump" shape), then declines as remaining cash flows decrease. PFE at 97.5% shows worst-case exposure.

How to Read This

The **hump-shaped** EE profile is characteristic of interest rate swaps: early in the trade, few cash flows have been exchanged so exposure is low. Mid-life, the swap has maximum sensitivity to rate changes. Near maturity, few cash flows remain, so exposure drops.
PFE 97.5% (red dashed) shows the exposure that would only be exceeded 2.5% of the time — this is the worst-case scenario used for credit limit setting. The peak PFE of \$90K on a \$1M notional means the counterparty could potentially owe you up to 9% of the notional under adverse rate moves.
CVA is computed by weighting EE by default probability at each time step: CVA $= (1 - R) \sum_i \mathrm{EE}(t_i) \cdot \Delta\mathrm{PD}(t_i) \cdot D(t_i)$.

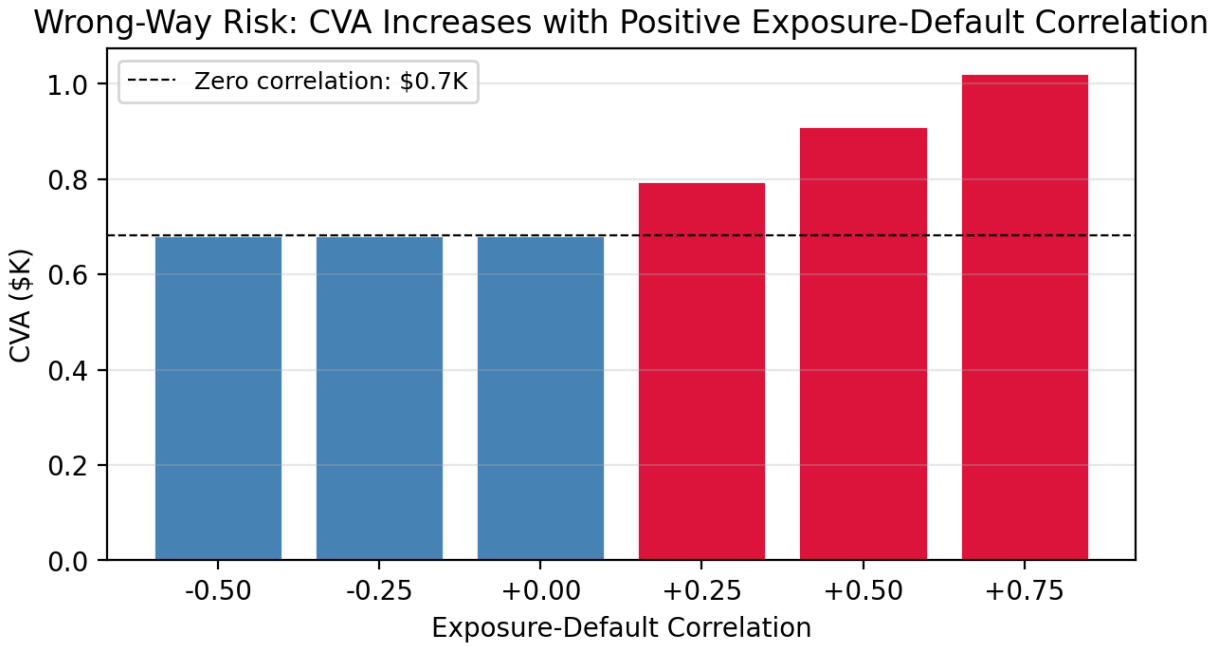

Figure 18: Wrong-way risk: CVA increases sharply when exposure and default probability are positively correlated. Negative correlation (right-way risk) reduces CVA.

How to Read This

Wrong-way risk is the nightmare scenario: your counterparty is most likely to default exactly when they owe you the most money. Example: you buy credit protection from a bank on mortgage securities — if mortgages crash, the bank's default probability spikes AND your claim against them grows.
The chart shows that at zero correlation, CVA is moderate. As **positive correlation** increases, CVA rises sharply — the expected loss increases because defaults cluster with high exposure periods. **Negative correlation** (right-way risk) has the opposite effect: defaults tend to occur when exposure is low, reducing CVA.
Basel III requires banks to identify and separately capitalize wrong-way risk exposures.

Technical Summary

#	Project	Tests	Key output
01	Portfolio Risk Dashboard	36	VaR/ES backtesting with Kupiec, Christoffersen, Basel
02	Monte Carlo Engine	17	Correlated simulation with variance reduction
03	Credit Scoring with ML	14	PD model with AUC = 0.86, SHAP explainability
04	Volatility Modeling	14	GARCH family + regime-switching
05	EVT for Tail Risk	12	GEV/GPD fitting, threshold selection, return levels
06	Copula Dependency	12	5 copula families, PIT, joint simulation
07	Stress Testing	14	DFAST, historical replay, reverse stress testing
08	Deep Hedging	13	Neural network hedging, CVaR optimization
09	CVA Counterparty Risk	14	Exposure profiles, netting, wrong-way risk
	Total	**132**	**16,000+ lines of tested, typed Python**

Stack: Python 3.11+, NumPy, SciPy, pandas, scikit-learn, XGBoost, arch, statsmodels, matplotlib.
Standards: SR 11-7 model governance, Basel III/IV FRTB alignment, full type hints, pytest with property-based testing.
Repository: `github.com/GonorAndres/risk-analyst` — MIT License.